

Assignment 8 – Builder Journal

Personal Productivity AI Agent

AI Engineering Fellowship 2026 – Week 3

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What I Built

I built a Personal Productivity AI Agent using LangGraph, OpenRouter, and Python. The agent manages tasks, notes, semantic search, work planning, and meeting action extraction through tool calling.

Most Difficult Technical Problem

The hardest challenge was implementing reliable human approval and integrating LangGraph routing with tool execution while keeping the agent state synchronized.

How I Solved It

I modularized the project into agent, tools, database, and vector store components, added an approval workflow, improved tool execution, and debugged state handling step by step.

Tool-Calling Errors Observed

Common issues included incorrect tool selection, missing arguments, delete-task routing, and malformed JSON responses from the LLM.

Agent Behavior That Surprised Me

The agent could often infer user intent correctly, but sometimes selected an incorrect tool for ambiguous prompts, highlighting the importance of prompt engineering.

What Failed During Testing

Delete-task synchronization, approval edge cases, and malformed responses required multiple debugging iterations before becoming stable.

What I Would Redesign

I would improve tool schemas with Pydantic validation, strengthen memory handling, and separate business logic further from controller logic.

What I Learned About Agent Reliability

Reliable AI agents require strong prompts, validation, approval checks, structured outputs, logging, and robust error handling rather than relying only on the LLM.

Goals for Week 4

Improve deployment, increase automated test coverage, optimize prompts, add richer session memory, and enhance UI and observability.